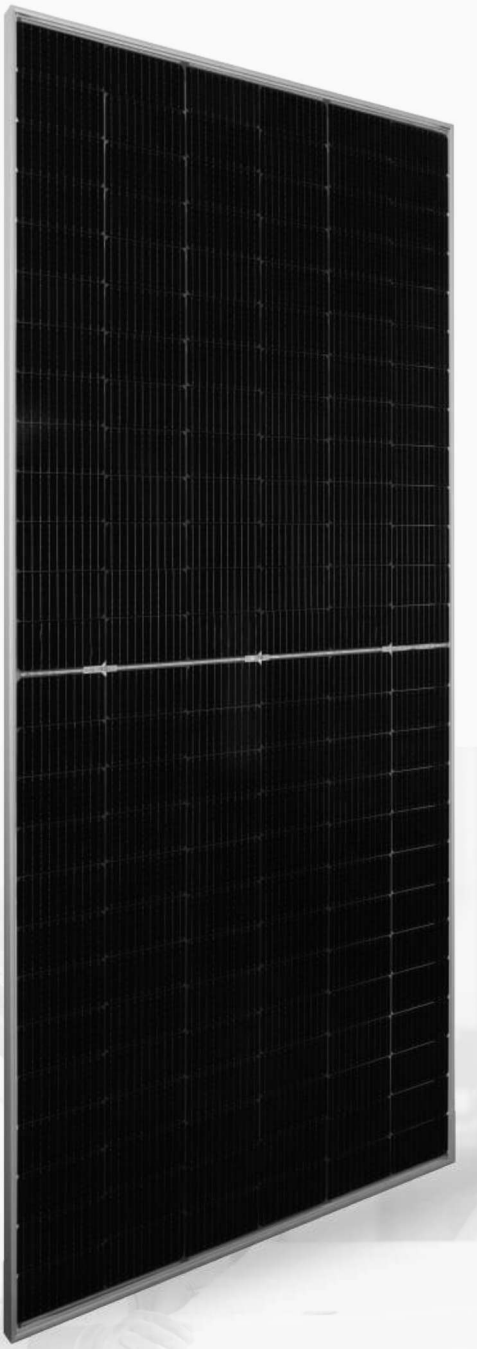
TP SOLAR LIMITED

TP600LG10B (Glass – Glass)

MONO PERC BIFACIAL SOLAR MODULE



Illuminating a Sustainable Future

Made in India within a state-of-the-art module production facility.

Low risk of failure - A warranty claim rate of under **0.07%.**

Certified to withstand snow loads of up to **5400Pa.**

Industry leading heavy-duty frame **6005-T6** grade.

PID resistant module.

IP68 rated junction box.

IEC 61215
IEC 61730
Regular Production Surveillance
www.tuv.com
ID 1111283735

ISO 9001:2015
BUREAU VERITAS
Certification

R-61004146

12 YEARS WARRANTY Product warranty on material and workmanship.

30 YEARS WARRANTY Linear power output warranty

Warranties are subject to the terms and conditions as per the Tata Power Solar warranty document.
Certification may vary by module type.

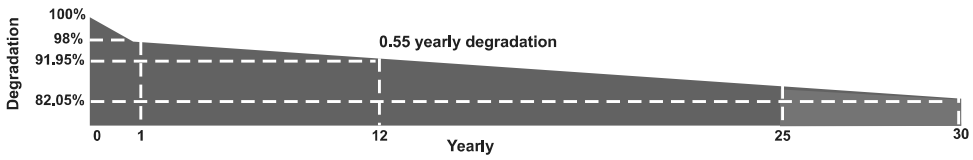
TP SOLAR LIMITED

21.54% Max Module Efficiency

600Wp Max Power Output

0 ~ +5 Power Tolerance

Yearly Degradation Chart



Electrical Parameters at STC

Maximum power - Pmax (W)	555	560	565	570	575	580	585	590	595	600
Power tolerance (W)	0 ~ +5	0 ~ +5	0 ~ +5	0 ~ +5	0 ~ +5	0 ~ +5	0 ~ +5	0 ~ +5	0 ~ +5	0 ~ +5
Module efficiency (η%)	19.93	20.11	20.29	20.47	20.65	20.83	21.00	21.18	21.36	21.54
Voltage at maximum power - Vmp (V)	45.05	45.17	45.50	45.71	45.87	45.97	46.10	46.18	46.28	46.41
Current at maximum power - Imp (A)	12.32	12.40	12.42	12.47	12.54	12.62	12.69	12.78	12.86	12.93
Open circuit voltage - Voc (V)	53.14	53.25	53.30	53.41	53.52	53.63	53.73	53.90	54.01	54.12
Short circuit current - Isc (A)	13.00	13.02	13.03	13.04	13.10	13.12	13.20	13.29	13.37	13.45

STC : Irradiance 1000 W/m², AM 1.5 and cell temperature 25°C

Electrical Parameters at 10% Bifacial gain

Maximum power - Pmax (W)	611	616	622	627	633	638	644	649	655	660
Voltage at maximum power - Vmp (V)	45.05	45.17	45.50	45.71	45.87	45.97	46.10	46.18	46.28	46.41
Current at maximum power - Imp (A)	13.55	13.64	13.66	13.72	13.79	13.88	13.96	14.05	14.14	14.22
Open circuit voltage - Voc (V)	53.14	53.25	53.30	53.41	53.52	53.63	53.73	53.90	54.01	54.12
Short circuit current - Isc (A)	14.36	14.46	14.58	14.67	14.77	14.87	14.97	15.05	15.15	15.25

Bifacial factor 70 ± 5% and bifacial power gain depends on the plant design, location and albedo

Electrical Parameters at NOCT

Maximum power - Pmax (W)	400	403	407	410	414	418	421	425	428	432
Voltage at maximum power - Vmp (V)	39.64	39.75	40.04	40.22	40.37	40.45	40.57	40.64	40.73	40.84
Current at maximum power - Imp (A)	10.08	10.14	10.16	10.20	10.26	10.32	10.38	10.45	10.52	10.58
Open circuit voltage - Voc (V)	46.76	46.86	46.90	47.00	47.09	47.19	47.29	47.43	47.53	47.62
Short circuit current - Isc (A)	10.65	10.66	10.67	10.68	10.73	10.75	10.81	10.89	10.95	11.01

NOCT : Irradiance 800 W/m², AM 1.5, Ambient temperature 20°C and Wind speed 1m/s

Temperature Coefficient Characteristics

NOCT(°C)	45±1
Temperature Coefficient of Pmax (% / °C)	-0.35
Temperature coefficient of Voc (% / °C)	-0.28
Temperature coefficient of Isc (% / °C)	0.05

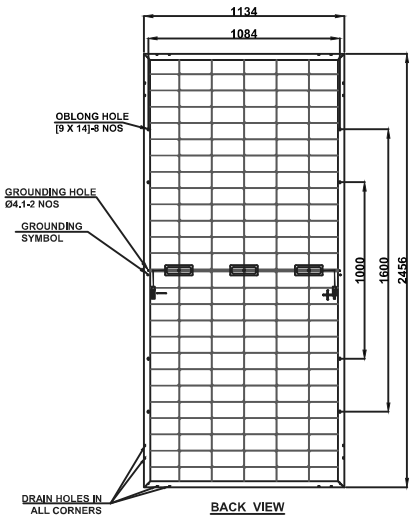
Operating Conditions

Maximum system voltage (V)	1500
Application class	A
Maximum series fuse rating (A)	25
Limited reverse current (A)	25
Operating temperature range (°C)	-40 to 85
Maximum static load (Wind/Snow) Pa	2400 /5400

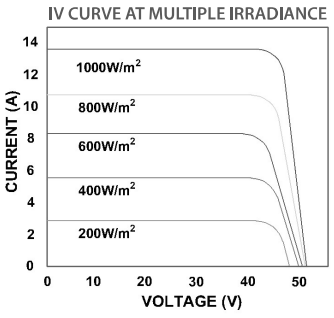
Module General Characteristics

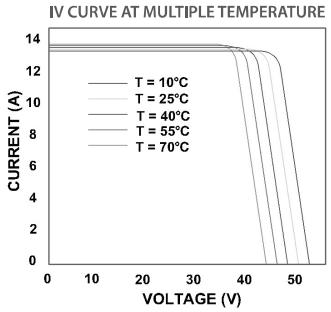
Module dimensions L X W X H(mm)	2456 x 1134 x 35
Module weight (approx) (Kg)	36±1
No of cells & size	156 Nos/[182 x91]mm
Frame materials	Anodized Aluminium
Front glass/ Back glass	2mm each side
Junction box	IP68
Cable connector	MC4/MC4 Compatible

Dimensions (mm)



IV Characteristics





Packaging Details

No. of Modules per pallet	31
No. of Boxes per 40ft Container	16
Box Weight (Kg)	1250
Box Dimensions L X W X H(mm)	2520 x 1145 x 1280

Testing and Tolerances

*Irradiance of 1000W/m², Spectrum of AM 1.5 and Cell temperature of 25 °C.
** Measurement tolerance on Pmax±3%
a] Tolerance for dimensions±3mm and Tolerance for cable length 0/+50mm.
b] Bifacial factor [70±5]%
c] BSTC: Bifacial power gain depends on the plant design and location
d] Frame : Lock Type-30mm/35mm
e] Box dimensions are subject to change & Box Dimension Tolerance ±10mm
Listed specifications are subject to change without prior notice.

भारतीय मानक ब्यूरो
(उपभोक्ता मामले, खाद्य एवं सार्वजनिक वितरण मंत्रालय, भारत सरकार)
BUREAU OF INDIAN STANDARDS
(Ministry of Consumer Affairs, Food & Public Distribution, Govt. of India)

मानक भवन, 9 बहादुर शाह जफ़र मार्ग, नई दिल्ली - 110002
Manak Bhavan, 9 Bahadur Shah Zafar Marg, New Delhi – 110002
दूरभाष/Phone: +91-11-23230856/2323010131/23233375/23239402
ई-मेल/E-mail: registration@bis.gov.in
वेबसाईट/Website: https://bis.gov.in/, https://www.crsbis.in/BIS/

Our Ref: REGISTRATION/CRS 2018-2889/R-62001090 Date:07-12-2022

Inclusion Id: 63272

Subject :Inclusion of Additional Model(s)

MANUFACTURING UNIT :	Tata Power Solar Systems Ltd M/S TATA POWER SOLAR SYSTEMS LTD, SY NO. 43P AND 44(P), ELECTRONIC CITY, 2ND STAGE, CONSISTING HOSUR ROAD, BANGALORE-560100 KARNATAKA,India-560100 dsmurty@tatapowersolar.com 9900545330	

Dear Sir,

1. This has reference to your request for inclusion of models of **"CRYSTALLINE SILICON TERRESTRIAL PHOTOVOLTAIC (PV) MODULES (Si wafer based)" as per IS 14286 : 2010/ IEC 61215 : 2005, IS/IEC 61730 (Part 1) : 2004 & IS/IEC 61730 (Part 2) : 2004 in Licence No. R-62001090** already granted to you which is valid upto 27-09-2024.

2. It is intimated that the additional Models as per details given below have been agreed to be included in your scope of Licence. **R-62001090 w.e.f. 07-12-2022:**

Product Category	CRYSTALLINE SILICON TERRESTRIAL PHOTOVOLTAIC (PV) MODULES (Si wafer based)
Product Name	Crystalline silicone terrestrial Photovoltaic(PV) Modules
IS No.	IS 14286 : 2010/ IEC 61215 : 2005, IS/IEC 61730 (Part 1) : 2004 & IS/IEC 61730 (Part 2) : 2004
Brand (As Declared by Manufacturer):	Tata Power Solar
Inclusion of Additional Models **(w.e.f. 07-12-2022)**	TP370SG10, TP375SG10, TP380SG10, TP385SG10, TP390SG10, TP395SG10, TP400SG10, TP405SG10, TP410SG10, TP415MG10, TP420MG10, TP425MG10, TP430MG10, TP435MG10, TP440MG10, TP445MG10, TP450MG10, TP455MG10, TP505HG10, TP510HG10, TP515HG10, TP520HG10, TP525HG10, TP530HG10, TP535HG10, TP540HG10, TP545HG10, TP550HG10, TP555LG10, TP560LG10, TP565LG10, TP570LG10, TP575LG10, TP580LG10, TP585LG10, TP590LG10, TP595LG10
Factory Address	M/S TATA POWER SOLAR SYSTEMS LTD, SY NO. 43P AND 44(P), ELECTRONIC CITY, 2ND STAGE, CONSISTING HOSUR ROAD, BANGALORE-560100 KARNATAKA,India-560100

3. Other terms and conditions of the licence shall remain same.

4. This letter is being issued with the approval of competent authority.

Kindly acknowledge receipt of this letter.

Thanking you,

Yours faithfully,
(Deepti Budiyal)
Granting Authority
Telfax : +91-11-23230856
E-mail: registration@bis.gov.in

Note: This is a system generated letter. Hence signature is not required.
To verify authentication of letter, kindly scan the QR code on this letter.

For details information on BIS, consult the e-BIS Portal (www.manakonline. in).
Please use BIS CARE APP for verification of ISI-marked goods and hallmarked gold jewellery.

भारतीय मानक ब्यूरो
(उपभोक्ता मामले, खाद्य एवं सार्वजनिक वितरण मंत्रालय, भारत सरकार)
BUREAU OF INDIAN STANDARDS
(Ministry of Consumer Affairs, Food & Public Distribution, Govt. of India)

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दूरभाष/Phone: +91-11-23230856/2323010131/23233375/23239402
ई–मेल/E-mail: registration@bis.gov.in
वेबसाईट/Website: https://bis.gov.in/, https://www.crsbis.in/BIS/

Our Ref: REGISTRATION /CRS-2023-4667/R-61004146

Dated: 2025-07-15 02:15:39

RENEWAL ID : 34656

Subject : RENEWAL OF LICENCE R-61004146 AS PER IS 14286 : 2010/ IEC 61215 : 2005, IS/IEC 61730 (PART 1) : 2004 & IS/IEC 61730 (PART 2) : 2004

TP Solar Limited
PLOT NO. A109, NEAR ELCOT ROAD, TP SOLAR LIMITED, SIPCOT ROAD AND OSR PARK GANGAIKONDAN ROAD, SIPCOT INDUSTRIAL PARK, GANGAIKONDAN INDUSTRIAL PARK, TIRUNELVELI,
TAMIL NADU, India, 627352

Dear Sir/Madam,

With reference to your online application dated 15-07-2025 for renewal of the above mentioned licence; this is to inform you that the same has been renewed from **10-10-2025** to **09-10-2027**.

It may be noted that the said licence granted under clause (b) of sub section (2) of section 13 of the Act shall *expire* at the end of the period for which it is granted unless renewed or its renewal is deferred. You are, therefore, requested to apply for next renewal to BIS within three months before the expiration of the licence.

Thanking you.

Yours faithfully,

Registration Department
Bureau of Indian Standards,
9, Bahadur Shah Zafar Marg,
New Delhi-110002.
Telfax : +91-11-23230856
E-mail: registration@bis.gov.in

Note: This is a system generated letter. Hence signature is not required.
To verify authentication of letter, kindly scan the QR code on this letter.

To Whom It May Concern

This is to certify that **M/s GUTS ELECTROMECH LTD**, having manufacturing units located at **Haridwar, Uttarakhand** and **Hyderabad, Telangana**, is a **wholly owned subsidiary of M/s V-GUARD INDUSTRIES LIMITED**, having its registered office at 42/962 (Old No. 33/2905 F), Vennala High School Road, Vennala, Kochi – 682028, Kerala, India.

We further certify that **GUTS ELECTROMECH LTD** is authorized by **V-GUARD INDUSTRIES LIMITED** to manufacture products under the brand name **"V-GUARD"**, and operates as an integral part of our in-house manufacturing operations.

We have **no objection** to **GUTS ELECTROMECH LTD, Haridwar and Hyderabad**, manufacturing and supplying products in the **V-GUARD** brand name for commercial and certification purposes.

This certificate is issued upon request and for necessary record and official use.

For **V-GUARD INDUSTRIES LIMITED**
Very truly yours,

Sign & Stamp

Name Sivasankaran A
GM - EMD

V-GUARD INDUSTRIES LTD. VENNALA COCHIN-28 VENNALA HIGH SCHOOL ROAD

Designation

V-GUARD INDUSTRIES LTD.
Regd. office 42/962,
Vennala, High School Road,
Vennala, Kochi - 682 028.
CIN: L31200KL1996PLC010010

P +91 484 433 5000, 200 5000
E mail@vguard.in
W www.vguard.in

Central Power Research Institute (CPRI), Bangalore

ULR No. : TC5452230SCLT0089F

DOC No. : CPRIBLRSCL23T0089
Telephone : 91 9448141980
FAX : 23601213
E-Mail : tbs@cpri.in
BO Code : NA

P.B.No. 8066, Sir.C V Raman Road, Sadashivanagar P O, Bengaluru, Bengaluru Urban, Karnataka, India - 560080

Test REPORT AS PER : IS/IEC 60947 : PART 3 (2012)

QR Code/Barcode : 134906CRS

REPORT NO : SC23HPI00014_1 DATE : 25 May, 2023

PART A. PARTICULARS OF SAMPLE SUBMITTED

a) Customer Name & Address	:	GUTS ELECTROMECH LTD. PLOT NO.2, SECTOR 3A, IIE, SIDCUL, HARIDWAR, NA, UTTARAKHAND, India - 249403
b) Nature of sample	:	-
c) Grade/Variety/Type/Class Size etc	:	NA
d) Declare values, if any	:	-
e) Batch No. & Date of Manufacture	:	/
f) Quantity	:	6
g) Date of Receipt	:	17 Jan, 2023
h) BIS Seal	:	Verified by Sample Cell
i) IO's Signature	:	Verified by Sample Cell
j) Any other Information / Expiry Date, If any	:	/
k) Date of Commencement of Testing	:	01 May, 2023
l) Date of Completion of Testing	:	16 May, 2023
m) Section Code	:	23E9D4AN
n) Section Report No.	:	23E9D4AN_1
o) Report Type	:	New
p) Reference Report No.	:	
q) Remarks	:	The sample tested complies with clauses 8.3.3, 8.3.4 & 8.3.6 of IS/IEC 60947-3:2012 (Reaffirmed-2018) for test sequences I, II & IV conducted on 415V 40A Uimp-4kVp AC-22A utilization category Double Pole Isolator.

RAMA DAS
OIC SAMPLE CELL
(Authorized Signatory)
Authorized on: 25 May, 2023 12:58 PM

1.

This is a Computer Generated Report.

CPRI

Annexure 1

Test Results

Test Report Number: CPRIBLRSCL23T0089 **Date: 28 April 2023**

QR Code	: 134906CRS
BIS Sample Code	: SC23HPI00014
Serial Number	: GELRND01234001
CPRI Sample code Number(s)	: SCL23S0253
Date of test	: 30 January, 2023

TEST SEQUENCE I: GENERAL PERFORMANCE CHARACTERISTICS

TEMPERATURE-RISE VERIFICATION [Subclause 8.3.3.1 of IS/IEC 60947-3 : 2012 (Reaffirmed 2018)]

Source
- Number of phases : Single
- Frequency : 50 Hz

Sample
- No. of poles : Two (All poles are connected in series)
- Mounting arrangement : Vertical

- Condition of the sample : Good

Test connections
- Cable size & length : 10 sq. mm. PVC insulated copper cable 1 m from source to sample, 1 m between poles & 1 m from sample to source

Rated current of 40 A was passed at ambient temperature of 24.8 °C and temperature rise recorded at steady state conditions.

Point of measurement on the sample	Temperature rise (K)	Limits (K)
Terminal 1	19.9	65
Terminal 2	19.8	65
Terminal 3	19.9	65
Terminal 4	19.2	65
Left side surface (Non-metallic)	7.6	50
Right side surface (Non-metallic)	7.4	50
Front side surface (Non-metallic)	3.3	40
Operating knob (Metallic)	2.0	15
Operating knob (Non-metallic)	2.1	25

(S. Shenbagarajan)
Test Engineer

CPRI

Annexure 2
Test Results

Test Report Number: CPRIBLRSCL23T0089 **Date: 28 April 2023**

QR Code	: 134906CRS
BIS Sample Code	: SC23HPI00014
Serial Number	: GELRND01234001
CPRI Sample code Number(s)	: SCL23S0253
Date of test	: 9 February & 13 March, 2023

TEST SEQUENCE I: GENERAL PERFORMANCE CHARACTERISTICS

TEST OF DIELECTRIC PROPERTIES [Subclause 8.3.3.2 of IS/IEC 60947-3 : 2012 (Reaffirmed 2018)]
General conditions for withstand voltage tests: The equipment was covered by aluminium foil and connected together to the mounting plate during the following tests.

Test conditions

Ambient temperature	: 25.7 °C
Relative Humidity	: 67.1 %
Rated impulse withstand voltage	: 4 kVp
Altitude corrected value	: 4.4 kVp & 5.6 kVp (for isolation)

a) IMPULSE WITHSTAND VOLTAGE TEST ON MAIN CIRCUITS

Type of test	Oscillogram number	Waveform parameters
Calibration*	CPRIBLRSCL23T0089S004	Peak = 4.46 kV, Rise time = 1.0 µs, Tail time = 47.3 µs
	CPRIBLRSCL23T0089S005	Peak = -4.44 kV, Rise time = 1.0 µs, Tail time = 48.1 µs

* 1.2 / 50 µs with a tolerance of ± 30% & ± 20% respectively
Five positive and five negative impulses were applied by connecting the above calibrated output to the test sample as follows:

Test procedure	Observation
Between all terminals of the main circuit connected together and the mounting plate with the contacts in all normal positions of operation.	Withstood No unintentional disruptive discharge occurred during the test.
Between each pole of the main circuit and the other poles connected to together and the mounting plate with the contacts in all normal positions of operation.	Withstood No unintentional disruptive discharge occurred during the test.

(S. Shenbagarajan)
Test Engineer